Take-Home Part I

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The Model and Data

Let $\{Y_i : i = 1, \ldots, n\}$ denote a set of response variables and let $\{x_i : i = 1, \ldots, n\}$ represent a corresponding set of non-decreasing covariate values. The model for the response variables is

$$Y_i = \mu_i + w_i \tag{1}$$

where

$$\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}. \tag{2}$$

In (2), $\alpha \geq 0$, $\beta \geq 0$, and $\gamma \geq 0$ are unknown parameters, and for $-1 < \theta < 1$,

$$w_i = \theta w_{i-1} + \epsilon_i, \tag{3}$$

where the ϵ_i are independent and identically distributed with $\epsilon_i \sim N(0, \sigma^2)$.

Q1

Consider a simplified model in which the w_i in (1) are iid $N(0, \sigma^2)$, i.e., a traditional additive error nonlinear regression model with constant variance. Fit this model to the project data using generalized least squares to estimate the parameters of the expectation function.

(a) and (b): Report GLS estimates of α, β, γ and the moment-based estimate of σ^2. Approximate 95% confidence intervals for α, β, γ.

Starting values are chosen by reasoning through $\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}$ from the outside in. The scatterplot shows the response still rising gradually near $x = 400$, suggesting slow exponent decay and thus a small γ; we set $\gamma_0 = 0.16$. At $x = 100$, the effective power on x is $\beta \cdot 100^{-0.16} \approx 0.48\beta$; a mild power-law rise (effective exponent ≈ 0.55) gives $\beta_0 \approx 0.55/0.48 \approx 1.2$. Finally, back-calculating from $y \approx 6.5$ near $x = 100$ gives $\alpha_0 \approx 0.5$.

Table 1: GLS estimates with approximate 95% confidence intervals.

		95% CI	
Parameter	Estimate	Lower	Upper
α	0.506	-0.331	1.343
β	1.222	0.211	2.233
γ	0.161	0.114	0.209

As noted in the solution to Question 1: The lower endpoint of the interval for $\hat{\alpha}$ is negative, which falls outside the scientifically meaningful ("desired") parameter space $\alpha \geq 0$, though it does not prevent μ_i from being defined. Also, the moment-based estimate of σ^2 is $\hat{\sigma}^2 = 1.015$.

(c) and (d): A plot with the estimated expectation function overlaid on the scatterplot along with an approximate 95% pointwise confidence band. A plot of standardized residuals against the covariate.

(c): As noted in the provided solution to Question 1: An approximate pointwise confidence band is constructed via the delta method, whose derivatives of μ_i with respect to (α, β, γ) are

$$\frac{\partial \mu_i}{\partial \alpha} = x_i^{\beta x_i^{-\gamma}}, \qquad \frac{\partial \mu_i}{\partial \beta} = \mu_i x_i^{-\gamma} \log x_i, \qquad \frac{\partial \mu_i}{\partial \gamma} = -\mu_i \beta (\log x_i)^2 x_i^{-\gamma}.$$

(d): Standardized residuals were computed using the hat-matrix correction $r_i/[\hat{\sigma}^2(1 - h_{i,i})]^{1/2}$, expression (9.40) of the Stat 520 notes, returned directly by `nonlin` via `fit1$stdres`. The solution sheet cites (9.37), the general plug-in form; (9.40) is its constant-variance simplification, obtained by substituting $\hat{\sigma}_i^2 = \hat{\sigma}^2$. The residual plot shows a good deal of remaining structure as a function of the covariate, consistent with Figure 3 of Sibbesen (1981). This pattern motivates the use of an autoregressive structure in the "regression equation error terms", i.e., the inclusion of an AR(1) small-scale component in the full model, treating the serial dependence across ordered covariate values analogously to serial correlation over time.

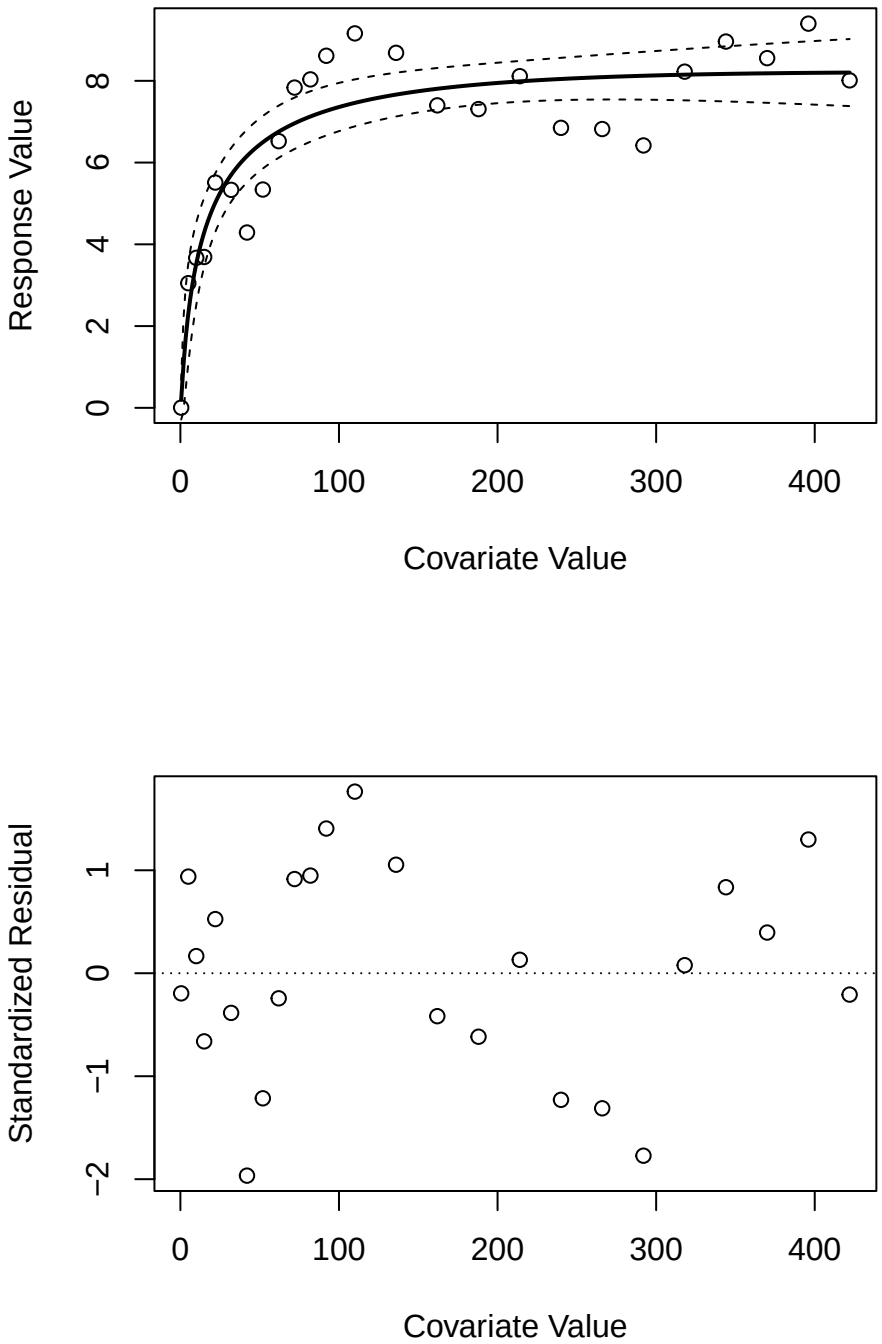

Figure 1: (Top) Fitted extended Freundlich curve with approximate 95% pointwise confidence band. (Bottom) Standardized residuals versus covariate.

Q2

Consider now just the portion of the model for w_i in (3). Suppose we were able to observe the w_i. Take $W_0 = 0$, and use without derivation that under second-order stationarity, $E(W_i) = 0$ and $E(W_i^2) = \sigma^2/(1-\theta^2)$ for all i.

(a) Show that the estimating equation for θ based on the naive estimating function

$$v_i(\mathbf{W}, \theta) = W_i - \theta W_{i-1}$$

is an unbiased estimating equation.

The naive estimating function is $v_i(\mathbf{W}, \theta) = W_i - \theta W_{i-1}$. Substituting $W_i = \theta W_{i-1} + \epsilon_i$ directly gives

$$E[v_i(W, \theta)] = E[\theta W_{i-1} + \epsilon_i - \theta W_{i-1}] = E[\epsilon_i] = 0,$$

since $\epsilon_i \sim N(0, \sigma^2)$ is independent of W_{i-1}. Because this holds for every i and all $-1 < \theta < 1$, the estimating equation $\sum_{i=2}^n v_i(\mathbf{W}, \theta) = 0$ is unbiased. Setting it to zero and solving gives

$$\tilde{\theta}_{\text{naive}} = \frac{\sum_{i=2}^n W_i}{\sum_{i=2}^n W_{i-1}}.$$

While this form is algebraically valid, the denominator $\sum W_{i-1}$ can be near zero, contributing to the numerical instability ($\tilde{\theta}_{\text{naive}} > 1$) observed when this estimating equation is applied to the Q1 residuals in Q2(d), *spoilers*.

(b) Show that the estimating equation for θ based on the optimal estimating function

$$v_i(\mathbf{W}, \theta) = W_{i-1}(W_i - \theta W_{i-1})$$

is an unbiased estimating equation.

The optimal estimating function is $v_i(\mathbf{W}, \theta) = W_{i-1}(W_i - \theta W_{i-1})$. Expanding the expectation and using the model $W_i = \theta W_{i-1} + \epsilon_i$,

$$E[W_{i-1} W_i] = E[W_{i-1}(\theta W_{i-1} + \epsilon_i)] = \theta E[W_{i-1}^2] + E[W_{i-1}\, \epsilon_i].$$

Because W_{i-1} is a function of $\epsilon_1, \ldots, \epsilon_{i-1}$ alone and ϵ_i is independent of those, $E[W_{i-1}\epsilon_i] = 0$.

Under the second-order stationarity assumed in the problem, which holds when $|\theta| < 1$, we have $E[W_{i-1}^2] = \sigma^2/(1-\theta^2)$,

$$E[v_i(\mathbf{W}, \theta)] = \theta \frac{\sigma^2}{1-\theta^2} - \theta \frac{\sigma^2}{1-\theta^2} = 0.$$

The estimating equation $\sum_{i=2}^n W_{i-1}(W_i - \theta W_{i-1}) = 0$ is therefore unbiased, and its solution is the lag-1 autocorrelation estimator

$$\tilde{\theta}_{\text{opt}} = \frac{\sum_{i=2}^n W_i W_{i-1}}{\sum_{i=2}^n W_{i-1}^2}.$$

This is called "optimal" in the sense of Godambe's criterion: Among linear estimating functions in W_{i-1}, multiplying the naive function by W_{i-1} maximizes the ratio $[E\{\partial v/\partial\theta\}]^2/\operatorname{var}\{v\}$ (see Ch. 8.3, the "measure of the lost efficiency of the solution to any estimating function"). Note that both estimating equations yield closed-form estimators, and the two estimators are in general numerically distinct.

(c) Develop a moment-based estimator for σ^2 that could be used with either estimating equation, and show that it is unbiased.

Because $\epsilon_i = W_i - \theta W_{i-1}$ and $\epsilon_i \sim N(0, \sigma^2)$, we have $E[\epsilon_i^2] = \sigma^2$ for all i. Plugging in $\tilde{\theta}$ from either estimating equation and using $W_0 = 0$ would motivate a possible estimator of the form

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (W_i - \tilde{\theta}\, W_{i-1})^2,$$

where $W_0 = 0$. Note: The $i = 1$ term contributes $W_1 - \tilde{\theta} W_0 = W_1$, which equals ϵ_1 by virtue of the boundary condition $W_0 = 0$; this is a boundary artifact rather than a general property of the innovation structure. To see that this estimator is approximately unbiased, note that at the true θ each term satisfies $E[(W_i - \theta W_{i-1})^2] = E[\epsilon_i^2] = \sigma^2$, so

$$E\left[\frac{1}{n} \sum_{i=1}^{n} (W_i - \theta W_{i-1})^2\right] = \sigma^2.$$

Once θ is replaced by $\tilde{\theta}$, however, strict unbiasedness no longer holds: $E[(W_i - \tilde{\theta} W_{i-1})^2] \neq \sigma^2$ exactly, because $\tilde{\theta}$ is itself a function of the data. The estimator is therefore only approximately unbiased, in the same spirit as s^2 in ordinary regression, with the $n-1$ denominator providing a degrees-of-freedom correction by analogy with the linear case.

(d) Apply both estimating equations using the raw residuals from Q1 as surrogate observations of w_i. Compute $\hat{\sigma}^2$ for each and compare all three estimates of σ^2 (naive EE, optimal EE, Q1 GLS). Are these estimators all estimating the same quantity?

Table 2: Comparison of $\hat{\theta}$ and $\hat{\sigma}^2$ estimates. Note that $\tilde{\theta}_{\text{naive}} > 1$ falls outside the desired parameter space $|\theta| < 1$.

Method	$\hat{\theta}$	$\hat{\sigma}^2$
Naive EE	1.0046	0.8559
Optimal EE	0.5422	0.6573
Q1 GLS	—	1.0149

Because the Q1 residuals $\hat{e}_i$ contain mean-estimation error, i.e., they are $y_i - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ rather than $w_i = y_i - \mu_i(\alpha, \beta, \gamma)$, the estimating equations applied here are only approximations to those derived in Q2(a) and (b). The two estimating equations yield different estimates of θ, though both target the same population parameter, that parameter being the AR(1) autocorrelation coefficient θ in model (3). The numerical difference between $\tilde{\theta}_{\text{naive}} = 1.005$ and $\tilde{\theta}_{\text{opt}} = 0.542$ reflects estimation error rather than any difference in what the equations are targetting, with the optimal estimate being the more reliable of the two by Godambe's criterion.

More importantly, the *three* $\hat{\sigma}^2$ values are **not** all estimating the same quantity. The Q1 GLS estimate was obtained under the assumption that the w_i are iid, so it targets $\text{Var}(w_i)$; because the true model is AR(1), or at least we are motivated to believe it is AR(1), $\text{Var}(w_i) = \sigma^2/(1-\theta^2)$. However, the Q1 GLS estimator itself has no knowledge of that structure. In contrast, the two EE-based estimates are obtained by squaring residuals of the form $W_i - \tilde{\theta} W_{i-1} \approx \epsilon_i$, so they target the innovation variance σ^2. Because the AR(1) process has positive autocorrelation ($\theta > 0$ appears typical for sorption data), the marginal variance

exceeds the innovation variance, and we expect $\hat{\sigma}^2_{\text{Q1}} > \hat{\sigma}^2_{\text{naive}}$ and $\hat{\sigma}^2_{\text{Q1}} > \hat{\sigma}^2_{\text{opt}}$. The two innovation-variance estimates both target σ^2 but are not expected to be numerically close: $\tilde{\theta}_{\text{naive}} = 1.005$ is inadmissible and the resulting squared innovations are distorted, so $\hat{\sigma}^2_{\text{naive}}$ and $\hat{\sigma}^2_{\text{opt}}$ can differ substantially even though they share a common target.

Q3

Assume $x_0 = 0$ and $w_0 = 0$ (based on Q1, Q2). Derive the full log likelihood for the model of (1), (2), and (3), and maximize it to find simultaneous estimates of α, β, γ, θ, and σ^2.

(a) An expression for the log likelihood.

Given $w_0 = 0$, the residuals $w_i = Y_i - \mu_i$ satisfy $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. The likelihood derived here is a *conditional* likelihood, conditioning on $w_0 = 0$ rather than treating W_1 as drawn from the stationary marginal distribution $N(0, \sigma^2/(1-\theta^2))$. Given this conditioning, each innovation $\epsilon_i = w_i - \theta w_{i-1}$ is independent, so the joint density of $(Y_1, \ldots, Y_n)$ may be written as a product of n normal densities. This approximation is adequate when $|\theta|$ is well away from 1, but becomes less accurate near the stationarity boundary, where the marginal and conditional likelihoods can differ meaningfully.

Under these conditions, the log likelihood may be written

$$\ell(\alpha, \beta, \gamma, \theta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(w_i - \theta w_{i-1}\big)^2,$$

where $w_i = Y_i - \mu_i = Y_i - \alpha x_i^{\beta x_i^{-\gamma}}$ and $w_0 = 0$.

(b) A description of how you maximized the log likelihood. Note that Q1 estimates provide natural starting values and $\theta_0 = 0$; `optim` with `method = "BFGS"` is recommended over Nelder-Mead for five parameters.

Wald

The log likelihood was maximized using `optim` with `method = "BFGS"`. Starting values for (α, β, γ) were taken from the Q1 GLS fit; $\theta_0 = 0$ and $\sigma_0^2 = \hat{\sigma}^2_{\text{Q1}}$ completed the five-parameter starting vector. These are natural starting values: the Q1 fit is the $\theta = 0$ special case of the full model, placing the initial iterate near the expected solution. Because `optim` minimizes by default, we pass the negative log likelihood. The Hessian was then extracted by specifying `hessian = TRUE` for Wald inference in part (c).

Profile Method

Profile likelihood confidence intervals were constructed to complement the Wald intervals, as Wald intervals can perform poorly when the log-likelihood is asymmetric or the MLEs are near parameter boundaries—both plausible concerns in this application. For each parameter $\psi \in \{\alpha, \beta, \gamma, \theta, \sigma^2\}$, the profile log-likelihood is

$$\ell_p(\psi) = \max_{\boldsymbol{\lambda}} \ \ell(\psi, \boldsymbol{\lambda}),$$

where $\boldsymbol{\lambda}$ denotes the remaining four parameters. At each point of a 60-point grid spanning $\hat{\psi} \pm 12$ Wald standard errors for (α, β, γ) and $\hat{\psi} \pm 4$ Wald standard errors for (θ, σ^2)—the wider range for the regression

parameters reflecting the asymmetric likelihood surface and the need to capture the $\alpha > 0$ boundary region—where the Wald SE is taken from the observed information matrix at the MLE, truncated to the parameter's natural support, the inner maximization over $\boldsymbol{\lambda}$ was performed using `optim` with `method = "BFGS"`, initialized at the full MLE. Results were confirmed with Nelder-Mead; BFGS was retained.

The 95% confidence interval for ψ is the set of grid values satisfying

$$\ell_p(\psi) \geq \hat{\ell} - \tfrac{1}{2}\chi^2_{0.95,\,1},$$

with endpoints obtained by linear interpolation at the threshold crossing. The choice to profile over single parameters is one of convenience; profiling over pairs or higher-dimensional subsets is equally valid but produced negligible differences in preliminary analysis.

(c) The MLEs, 95% confidence intervals (stating whether Wald or profile), and the maximized log likelihood value.

Wald

Wald confidence intervals were constructed from the observed information matrix — the negative Hessian of ℓ evaluated at the MLE, returned directly by `optim` via `hessian = TRUE` — used in place of the expected Fisher information $\mathcal{I}(\boldsymbol{\beta}) = -E(\nabla^2\ell)$. These are asymptotically equivalent under standard regularity conditions, though the distinction is worth noting: with $n = 23$ and $p = 5$, these intervals rely heavily on asymptotic normality and should be interpreted with caution.

Profile Method

Profile 95% confidence intervals are reported alongside the Wald intervals in Table 3. The most consequential difference is for α: the Wald lower bound of -0.4834 is negative and therefore inadmissible under the model, which requires $\alpha \geq 0$. The profile lower bound of 0.0801 correctly respects this constraint, reflecting the asymmetry of the likelihood surface near the boundary. Right-skew is also evident for σ^2. Where the two interval types agree closely — as for γ and θ — the likelihood surface is approximately quadratic and the Wald approximation is adequate. Given the inadmissibility of the Wald interval for α, profile confidence intervals are used as the basis for all subsequent inference.

Profile log-likelihood maxima matched $\hat{\ell} = -29.598$ to within 0.001 across all parameters, confirming that the inner BFGS optimization successfully recovered the MLE at each fixed value.

The profile upper bound for θ of 0.9151 warrants note: it lies close to the stationarity boundary $|\theta| = 1$, meaning the data are consistent with near-unit-root serial dependence. Under near-unit-root conditions, the marginal variance $\sigma^2/(1-\theta^2)$ becomes large and poorly estimated, and the conditional likelihood approximation based on $w_0 = 0$ may be less accurate than asymptotic theory suggests. This is a limitation that cannot be resolved at $n = 23$.

Table 3: Wald and profile 95% confidence intervals at the MLE. Maximized log likelihood: -29.5981.

		Wald 95% CI		Profile 95% CI	
Parameter	MLE	Lower	Upper	Lower	Upper
α	0.7395	-0.4834	1.9625	0.0801	2.2498
β	1.0283	0.0275	2.0291	0.3854	2.0480
γ	0.1574	0.0929	0.2219	0.0682	0.2188
θ	0.5598	0.2203	0.8993	0.2102	0.9152
σ^2	0.6250	0.2785	0.9714	0.3763	1.1510

(d) A plot of the data with the estimated expectation function overlaid along with an approximate 95% pointwise confidence band.

The confidence band for $\mu_i = \alpha x_i^{\beta x_i^{-\gamma}}$ uses the delta method with the 3×3 submatrix of $\widehat{\text{Cov}}$ corresponding to (α, β, γ). The derivatives are the same as in Q1.

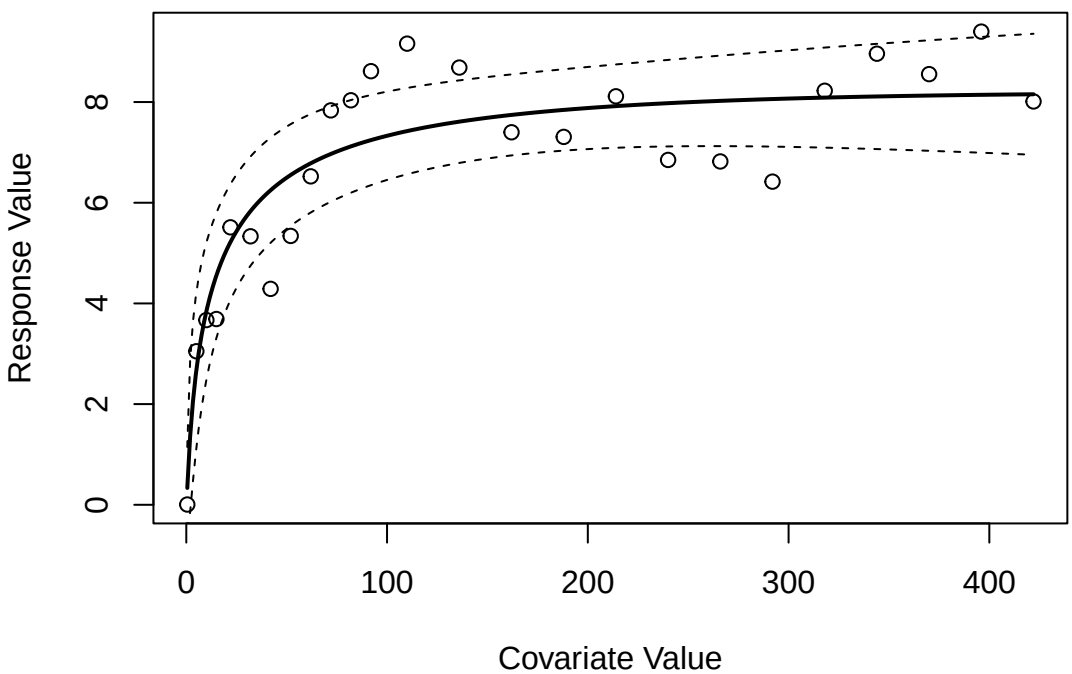

Figure 2: Fitted extended Freundlich curve with approximate 95% pointwise confidence band via the delta method. Scalar parameter CIs in (c) use profile intervals; the band uses the Wald delta method as profile bands for functions of multiple parameters are not standard.

(e) A comparison of the large-scale component (α, β, γ) to Q1: point estimates, interval estimates, estimated correlations among parameter estimators, and confidence band widths. Discuss any differences.

Table 4: Point estimates and 95% CI widths by method (Q1 Wald, Q3 profile). Mean confidence band half-width: Q1 = 0.6225, Q3 = 0.9666.

	Point Estimate		CI Width	
Parameter	Q1 GLS	Q3 MLE	Q1 GLS	Q3 Profile
α	0.5063	0.7395	1.6737	2.1697
β	1.2220	1.0283	2.0214	1.6625
γ	0.1614	0.1574	0.0944	0.1506

Table 5: Estimated correlation matrix for (α, β, γ): Q1 GLS.

	α	β	γ
α	1.000	-0.987	-0.837
β	-0.987	1.000	0.914
γ	-0.837	0.914	1.000

Table 6: Estimated correlation matrix for (α, β, γ): Q3 MLE.

	α	β	γ
α	1.000	-0.968	-0.685
β	-0.968	1.000	0.843
γ	-0.685	0.843	1.000

The point estimates of (α, β, γ) shift noticeably between Q1 and Q3 — most visibly $\hat{\alpha}$ moving from 0.5063 to 0.7395 — because jointly estimating θ reweights the observations according to the AR(1) correlation structure rather than treating them as independent. When correlation parameters are known, GLS with $\Sigma^{-1}(\theta)$ as a fixed weight matrix improves efficiency over the iid estimator. When they must be estimated jointly, however, the additional uncertainty in θ and σ^2 propagates into the regression parameters and can widen confidence intervals.

The profile CI widths under Q3 are wider than Q1 Wald for α and γ, narrower for β — likely because the shift in $\hat{\beta}$ toward the interior of the likelihood ridge under Q3 places it in a region of higher curvature — and the mean confidence band half-width increases from 0.6225 to 0.9666. The inadmissible Wald lower bound of -0.4834 for α is replaced by a profile lower bound of 0.0801, correctly respecting the $\alpha > 0$ constraint. This reflects the additional uncertainty introduced by estimating θ and σ^2 jointly with the regression parameters: the likelihood surface is more diffuse in the (α, β, γ) directions once the correlated error structure is acknowledged, and the Q1 covariance matrix, which ignored this coupling, was understating that uncertainty. It should also be noted that Table 4 compares Q1 Wald intervals to Q3 profile intervals — methodologically distinct constructions. Profile intervals are generally wider than Wald when the likelihood surface is asymmetric, so part of the observed width difference reflects the interval type rather than purely the model specification change.

Part of the observed width difference is also attributable to $\hat{\sigma}^2_{\text{Q1}} = 1.015 > \hat{\sigma}^2_{\text{MLE}} = 0.625$, which inflates Q1 standard errors; the Q3 intervals would be even wider if σ^2 were held equal across methods. The estimated correlations among the regression parameter estimators also shift under Q3, with $|\hat{\rho}_{\alpha,\beta}| \approx 0.97$ under both methods indicating that the likelihood surface is extremely flat along the (α, β) ridge — meaning individual parameter estimates are highly sensitive to small perturbations regardless of the fitting method, and the wide confidence intervals are intrinsic to the model structure rather than an artifact of the estimation procedure.

(f) A comparison of the small-scale component (θ, σ^2) to the optimal estimating equation results from Q2. Two estimates of θ and three estimates of σ^2 are available (Q3 MLE, Q1 GLS, Q2 moment-based). Discuss any differences.

Table 7: Comparison of small-scale parameter estimates.

Quantity	Q2 Optimal EE	Q3 MLE
θ	0.5422	0.5598
σ^2	0.6573	0.6250

Table 8: All estimates of σ^2 across methods.

Estimator	$\hat{\sigma}^2$
Q1 GLS (marginal variance)	1.0149
Q2 moment-based	0.6573
Q3 MLE	0.6250

The two estimates of θ agree closely: the Q2 optimal estimating equation yields $\hat{\theta} = 0.5422$ and the Q3 MLE yields $\hat{\theta} = 0.5598$. Both target the same parameter but differ in how they handle the regression function. The Q2 estimate used raw residuals from the misspecified iid model of Q1 as surrogates for w_i, introducing noise from the first-stage estimation; the Q3 MLE estimates θ simultaneously with (α, β, γ), which should reduce that bias. The small discrepancy is consistent with this explanation.

For σ^2, the three estimators target different quantities, as established in Q2(d): the Q1 GLS estimate of 1.0149 targets the marginal variance $\sigma^2/(1-\theta^2)$ and is therefore expected to be the largest when $\theta > 0$. Both the Q2 moment-based estimate (0.6573) and the Q3 MLE (0.6250) target the innovation variance σ^2, and are indeed substantially smaller. The Q2 version inherits approximation error from using Q1 residuals instead of true w_i values, while the Q3 MLE is obtained from the correctly specified joint model; the remaining discrepancy of 0.0323 between them is attributable to that first-stage approximation.

Q4

Assume $x_0 = 0$ and $w_0 = 0$. Using the events $A_i = (Y_i = y_i \mid Y_{i-1} = y_{i-1})$ for $i = 2, \ldots, n$, derive a pairwise conditional log composite likelihood and fit the model. Report the same components as in Q3.

Additionally, discuss which asymptotic context for composite likelihood is most appropriate given $n = 23$. Address potential issues with estimating the H and J matrices in the Godambe information; compute them if possible and use them to construct interval estimates for the regression function parameters.

(a) Expression for the Log Composite Likelihood

Given the model $w_i = Y_i - \mu_i$ and the AR(1) structure $w_i = \theta w_{i-1} + \epsilon_i$ with $\epsilon_i \sim N(0, \sigma^2)$. Since $w_i \mid w_{i-1} \sim N(\theta w_{i-1}, \sigma^2)$, the conditional distribution of Y_i given $Y_{i-1} = y_{i-1}$ is

$$Y_i \mid Y_{i-1} = y_{i-1} \sim N\big(\mu_i + \theta(y_{i-1} - \mu_{i-1}),\ \sigma^2\big).$$

The corresponding conditional density is

$$f(y_i \mid y_{i-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\big[(y_i - \mu_i) - \theta(y_{i-1} - \mu_{i-1})\big]^2}{2\sigma^2} \right).$$

Letting $\epsilon_i = (y_i - \mu_i) - \theta(y_{i-1} - \mu_{i-1})$ and applying log simplification rules, each conditional log-likelihood piece is

$$\ell_i = \log f(y_i \mid y_{i-1}) = \log\Big[(2\pi\sigma^2)^{-1/2}\Big] + \log\left[\exp\left(-\frac{\epsilon_i^2}{2\sigma^2}\right)\right] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\sigma^2 - \frac{\epsilon_i^2}{2\sigma^2}.$$

Summing over $i = 2, \ldots, n$, the constants $-\frac{1}{2}\log(2\pi)$ and $-\frac{1}{2}\log\sigma^2$ each accumulate $n-1$ times and σ^2 factors out of the sum, giving the pairwise conditional log composite likelihood

$$\ell_C = \sum_{i=2}^{n} \ell_i = \sum_{i=2}^{n} \left[-\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\sigma^2 - \frac{\epsilon_i^2}{2\sigma^2} \right] = -\frac{n-1}{2}\log(2\pi) - \frac{n-1}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=2}^{n}\epsilon_i^2.$$

The composite likelihood thus conditions on Y_1 (i.e., sums from $i = 2$), whereas the full likelihood in Q3 used $w_0 = 0$ to include $i = 1$, so the two differ by exactly one term.

(b) Maximization

The log composite likelihood was maximized using `optim` with `method = "BFGS"`, using Q1 GLS estimates as starting values for (α, β, γ) and $\theta_0 = 0$, $\sigma_0^2 = \hat{\sigma}^2_{\text{Q1}}$.

(c) MCLEs, Confidence Intervals, and Maximized Log Composite

Two sets of 95% confidence intervals are reported in Table~9: Godambe Wald (sandwich covariance $\hat{G}^{-1}$) and Godambe-adjusted profile (preferred). the adjusted profile threshold requires the λ_ψ scaling correction derived below; the adjusted profile intervals are preferred as they correctly respect the $\alpha > 0$ boundary and account for likelihood asymmetry.

Asymptotic Context and Godambe Information

This problem falls under Asymptotic Context II (Ch. 9.3.2): a single realization of a length-$n = 23$ dependent sequence, not independent replicates of a multivariate vector. Neither formal asymptotic result from Ch. 9 applies directly. Context I requires iid replicates, which are unavailable. The Comets–Janzura result for Context II requires translation-invariant conditional specification, which fails here because μ_i varies with the covariate. The question is then whether computing empirical analogs of H and J has any theoretical grounding, or whether it is purely heuristic — and the honest answer is that the grounding is weaker than in either formal context. What follows provides the best available justification by drawing on Geyer's (2020) lecture notes on unbiased estimating equations and Geyer's (2013) paper on likelihood asymptotics without the LLN, CLT, or sample size going to infinity, while acknowledging that the bootstrap remains the more reliable alternative at $n = 23$.

The justification lies in the structure of the AR(1) model itself. Each piece of the conditional composite likelihood conditions on Y_{i-1}, so the residual entering ℓ_i is exactly $\epsilon_i = e_i \sim$ iid $N(0, \sigma^2)$. The piece score $\nabla \ell_i$ is a function of e_i and Y_{i-1}, and because $e_i \perp Y_{i-1}$ by the AR(1) structure,

$$E[\nabla \ell_i \mid Y_{i-1}] = 0.$$

The sequence $\{\nabla \ell_i(\boldsymbol{\psi}_0)\}$ evaluated at the *true* parameter $\boldsymbol{\psi}_0$ is therefore a martingale difference sequence (MDS) — a concept not covered in this course, but one that carries a CLT and LLN under mild regularity conditions, without requiring iid replicates and without requiring stationarity of the Y_i themselves. At the *estimated* $\hat{\boldsymbol{\psi}}$, the MDS property holds only approximately; the standard justification appeals to continuity of the score and consistency of the MCLE, which together imply that $\nabla \ell_i(\hat{\boldsymbol{\psi}}) \approx \nabla \ell_i(\boldsymbol{\psi}_0) + o_p(1)$ termwise, preserving the asymptotic structure. In the language of Geyer (2020), the composite score is an unbiased estimating function whose asymptotic variance takes the sandwich form $U^{-1}V(U^{-1})^T$, which is inverse Godambe information. The MDS property is precisely what makes the natural estimators of U and V — namely $\hat{H}$ and $\hat{J}$ — consistent as estimators of those matrices (conditions (8) and (9) of Geyer (2020), which are the requirements for the sandwich covariance to be valid).

Geyer (2013) offers a complementary perspective: approximate normality of the MCLE follows if the composite log likelihood is approximately quadratic with Hessian approximately invariant in law, with no iid or stationarity requirement. Whether that holds here is an empirical question about this particular surface rather than a theoretical guarantee — Geyer explicitly identifies the AR(1) likelihood as a non-example of LAN/LAMN (Geyer, 2013, Section 2.3).

With $n = 23$ observations and five parameters, $\hat{H}$ and $\hat{J}$ should be interpreted cautiously regardless; the finite-sample performance of both matrices is uncertain. The MDS argument provides the theoretical grounding that makes computing the Godambe intervals meaningful rather than purely heuristic, and guarantees a limiting distribution for any bootstrap procedure to approximate. As noted in Ch. 9.3.3, the parametric bootstrap remains the more reliable alternative in practice.

$\hat{H}$, $\hat{J}$, and Godambe-Based Intervals

With this justification in hand, the empirical analogs are computed as follows. $\hat{H}$ is taken as the negative Hessian of ℓ_C evaluated at the MCLE, returned directly by `optim`. $\hat{J}$ is the sum of outer products of individual piece score vectors $\nabla \ell_i$, approximated by numerical central differences at the MCLE. This outer-product form parallels the Context I estimator in Ch. 9.3.1, but its justification here is the MDS structure described above rather than averaging over iid replicates; the Comets–Janzura neighborhood-sum form (Ch. 9, eq. 9.34) does not apply and is not used.

Examining Tables 10 and 11, the $(3, 3)$ elements of both $\hat{H}$ (38593.78) and $\hat{J}$ (40035.62) are very large relative to all others, reflecting high curvature in the γ direction and indicating that γ is the most precisely determined parameter locally. The large negative off-diagonal entries in rows/columns 1–3 and 2–3 are consistent with the high (α, γ) and (β, γ) correlations seen in Table 15, and confirm that the identifiability challenges in this model are concentrated in the (α, β, γ) block.

Profile Composite Likelihoods

The Godambe Wald intervals in Table~9 do not respect parameter boundaries and inherit the symmetric normal approximation. Profile intervals address both limitations, as in Q3, but the composite likelihood ratio statistic requires a Godambe-based correction before comparison to a χ^2_1 threshold.

Profile composite likelihood intervals are therefore constructed analogously to Q3, with one modification. The composite likelihood ratio statistic

$$W_C(\psi_0) = 2\big[\ell_C(\hat{\psi}) - \ell_C(\psi_0,\, \hat{\boldsymbol{\lambda}}_{\psi_0})\big]$$

is not asymptotically χ^2_1 in general; it converges to $\lambda_\psi \chi^2_1$, where the scalar correction

$$\lambda_\psi = \frac{(\hat{G}^{-1})_{\psi,\psi}}{(\hat{H}^{-1})_{\psi,\psi}} = \frac{\text{Godambe variance for } \psi}{\text{Wald variance for } \psi}$$

is the ratio of the Godambe to Wald marginal variance for parameter ψ, evaluated at the MCLE. The adjusted profile threshold is therefore

$$\ell_C(\psi) \;\geq\; \hat{\ell}_C - \tfrac{1}{2}\,\lambda_\psi\,\chi^2_{0.95,\,1}.$$

Because $\hat{J} \approx \hat{H}$ numerically (Tables 10 and 11), $\lambda_\psi \approx 1$ for all parameters, so the correction has modest practical effect here. The adjusted profile intervals are reported alongside the Godambe Wald intervals in Table~9. Note that $\hat{H}$ and $\hat{J}$ are not available until after the Godambe computation; the profile grid is evaluated first and the threshold is applied retrospectively once λ_ψ is known.

The CIs in Table~9 reveal a consistent pattern across methods: α and β shift monotonically as the likelihood changes (Q1 Wald $\rightarrow$ Q3 Profile $\rightarrow$ Q4), while γ remains relatively stable. The Q4 Godambe intervals are wider than Q3 profile for α and β, reflecting the additional uncertainty from losing one observation in the pairwise conditional construction.

The adjusted profile lower bound for γ does not exist at the 95% level: the composite likelihood surface to the left of $\hat{\gamma} = 0.145$ remains above the adjusted threshold all the way to the parameter boundary, in contrast to the full likelihood of Q3, which descends below its threshold and yields a lower bound of 0.068. This is a direct consequence of the composite likelihood using only $n - 1 = 22$ pairs rather than all $n = 23$ observations — the reduction in information is just enough to render γ's lower bound undefined under the composite likelihood.

Because the (α, β, γ) block of the adjusted profile covariance is not well-defined when one marginal profile is unbounded, a δ-method confidence band cannot be constructed from it in the usual way. The Godambe Wald covariance $\hat{G}^{-1}$ is therefore used for the band, despite its inadmissible lower bound for α of -0.818.

Table 9: Godambe Wald and adjusted profile 95% confidence intervals. Maximized composite log likelihood: -28.7688.

		Godambe Wald 95% CI		Godambe Adj. Profile 95% CI	
Parameter	MCLE	Lower	Upper	Lower	Upper
α	1.0628	-0.8182	2.9437	0.2424	3.5539
β	0.8126	-0.2991	1.9242	0.1866	1.5941
γ	0.1448	0.0472	0.2425	NA	0.2056
θ	0.5752	0.2250	0.9254	0.2219	0.9549
σ^2	0.6437	0.4061	0.8814	0.4547	0.9540

Table 10: Estimated H matrix (negative Hessian of composite log likelihood at MCLE).

	α	β	γ	θ	σ^2
α	328.3120	831.6470	-3442.0950	-2.2842	0.0257
β	831.6470	2122.1757	-8903.9726	3.2073	-0.0161
γ	-3442.0950	-8903.9726	38593.7805	-61.7570	0.3338
θ	-2.2842	3.2073	-61.7570	36.4902	-0.0082
σ^2	0.0257	-0.0161	0.3338	-0.0082	28.9569

Table 11: Estimated J matrix (sum of outer score products at MCLE).

	α	β	γ	θ	σ^2
α	332.8818	851.2476	-3562.8526	-5.8426	-18.2499
β	851.2476	2184.9056	-9229.8893	-12.2185	-44.2914
γ	-3562.8526	-9229.8893	40035.6206	31.7402	153.9595
θ	-5.8426	-12.2185	31.7402	29.5949	1.7960
σ^2	-18.2499	-44.2914	153.9595	1.7960	12.3473

(d) Fitted Curve and 95% Pointwise Confidence Band

For scalar parameters, the Godambe-adjusted profile intervals (Table~9) are preferred. However, constructing a pointwise confidence band requires a well-defined joint covariance matrix over (α, β, γ) via the delta method, and this cannot be formed from the adjusted profile: the profile lower bound for γ does not exist at the 95% level, leaving the (α, β, γ) covariance block undefined. The Godambe Wald covariance $\hat{G}^{-1}$ is therefore used for the band~— not as an approximation to the profile, but as the only well-defined joint covariance available.

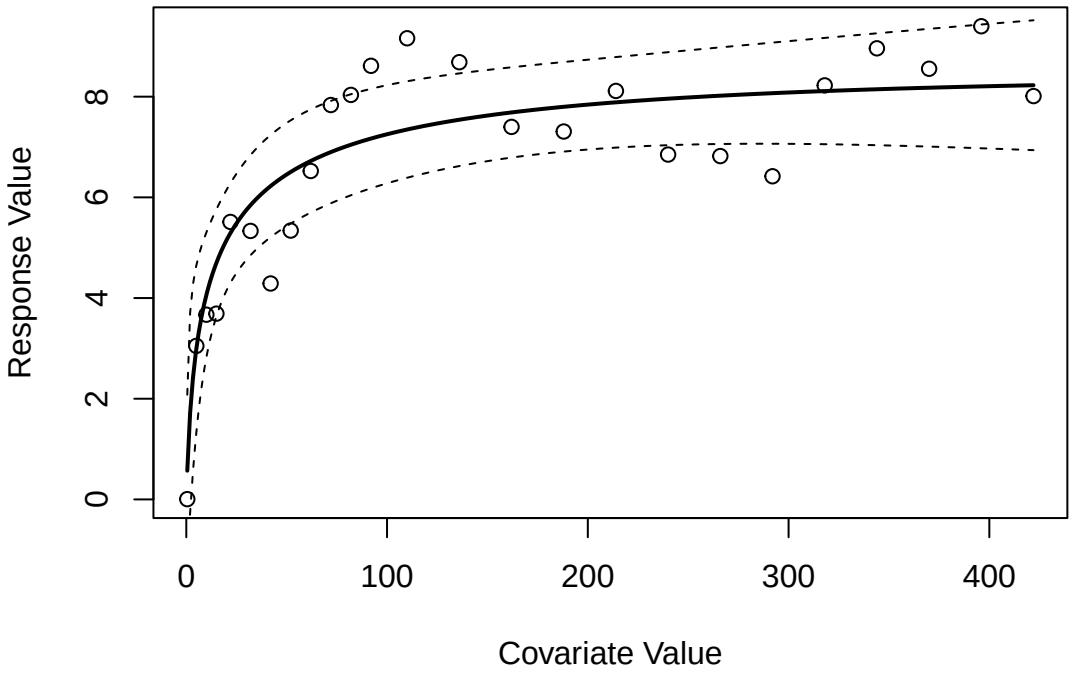

Figure 3: Fitted extended Freundlich curve with approximate 95% pointwise confidence band based on the Godambe covariance. Band uses the delta method with the (α, β, γ) block of $\hat{G}^{-1}$; see Q3(d) for the analogous Wald construction.

(e) Comparison of Large-Scale Component to Q1 and Q3

Table 12: Point estimates of large-scale parameters across methods.

Parameter	Q1 GLS	Q3 MLE	Q4 MCLE
α	0.5063	0.7395	1.0628
β	1.2220	1.0283	0.8126
γ	0.1614	0.1574	0.1448

Table 13: 95% CI widths for large-scale parameters across methods (Q1 Wald, Q3 profile, Q4 Wald and Godambe). Mean confidence band half-width (Q4): 1.0281.

			Q4	
Parameter	Q1 GLS	Q3 MLE	Q4 Wald	Q4 Godambe
α	1.6737	2.1697	5.4666	3.7620
β	2.0214	1.6625	3.0152	2.2234
γ	0.0944	0.1506	0.2292	0.1953

Table 14: Estimated correlation matrix for (α, β, γ): Q4 Godambe.

	α	β	γ
α	1.000	-0.980	-0.845
β	-0.980	1.000	0.932
γ	-0.845	0.932	1.000

The point estimates show a clear monotone progression across methods: $\hat{\alpha}$ increases from 0.506 (Q1) to 0.740 (Q3) to 1.063 (Q4), while $\hat{\beta}$ and $\hat{\gamma}$ decrease correspondingly. This reflects movement along the likelihood ridge defined by the near-collinearity among the large-scale parameters (Table 14: $\hat{\rho}_{\alpha,\beta} = -0.980$, $\hat{\rho}_{\alpha,\gamma} = -0.845$), not sampling noise. The direction of the progression is also structurally meaningful: Q1 ignores θ entirely, Q3 conditions on $w_0 = 0$ using all n terms, and Q4 conditions on Y_1 using only $n - 1$ terms, so each successive reduction in the likelihood contributes to the shift.

The CI widths in Table 13 tell a nuanced story. Q4 Wald intervals are the widest for every parameter — substantially so for α (5.467 vs 2.170 under Q3 profile) — because the inverse-Hessian covariance inherits the near-singular curvature of the five-parameter surface without the stabilizing contribution of the first observation. The Godambe-based intervals are uniformly narrower than Q4 Wald, because $\hat{J} \approx \hat{H}$ numerically (Tables 11 and 12), so the sandwich correction $\hat{H}^{-1}\hat{J}\hat{H}^{-1}$ deflates slightly relative to $\hat{H}^{-1}$ alone. With only $n - 1 = 22$ outer products contributing to $\hat{J}$, the agreement may partly reflect insufficient sample size to reliably distinguish the two matrices — though under correct model specification $J = H$ is expected asymptotically, so the approximate equality is at minimum consistent with an adequate composite likelihood specification.

The standardized residual plots (Figure 3) look visually similar across all three methods, but the comparison is asymmetric: the Q3 and Q4 innovation residuals ($\hat{\epsilon}_i = \hat{w}_i - \hat{\theta}\hat{w}_{i-1}$) have had the AR(1) dependence explicitly removed by construction, so their lack of trend confirms whitening rather than model adequacy. More notable is the mild cyclical pattern present in all three panels — positive residuals at low and high covariate values, with a slight dip in the mid-range — which points to a small remaining misspecification

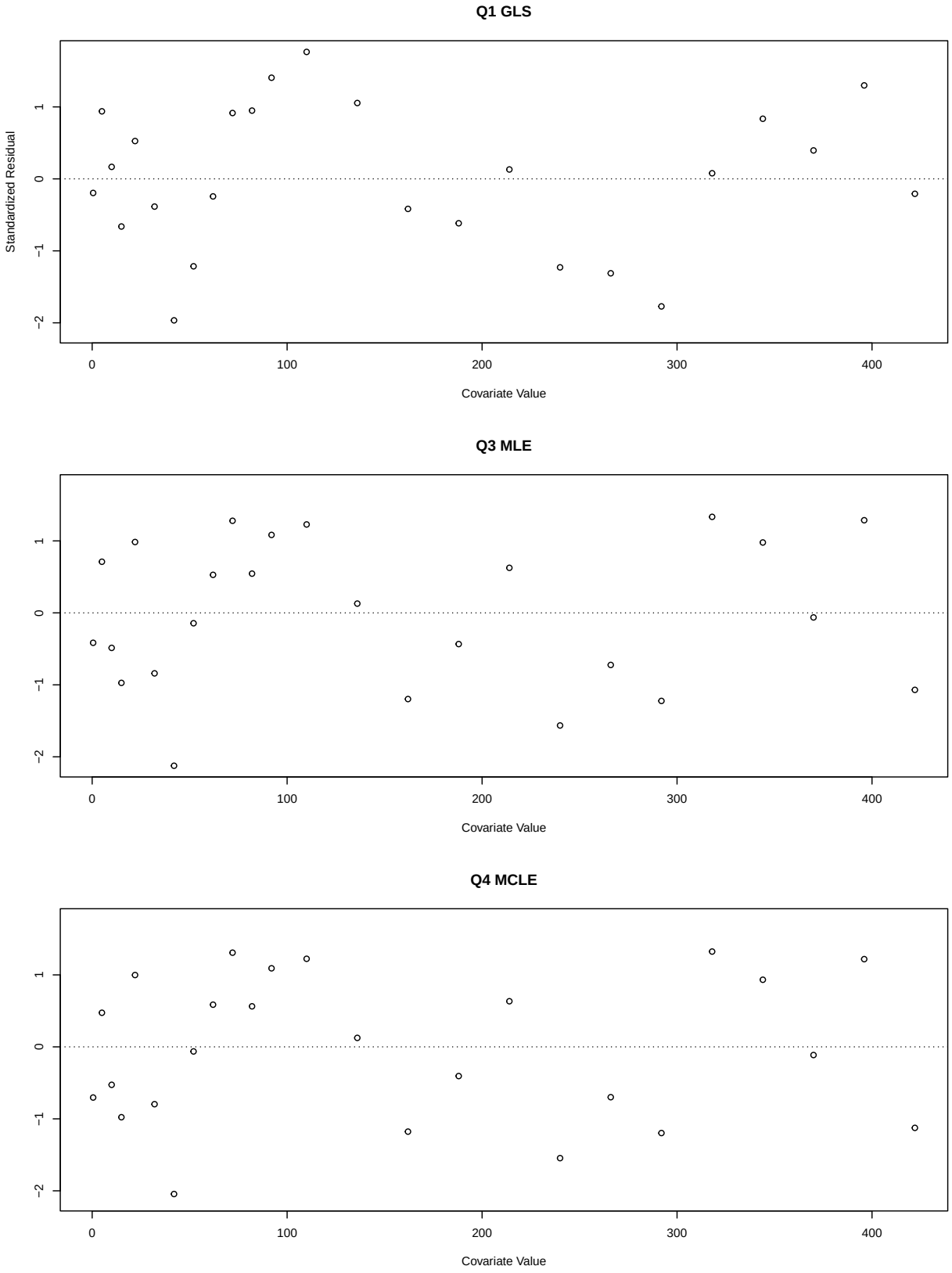

Figure 4: Standardized residuals versus covariate under Q1 GLS (left), Q3 MLE (center), and Q4 MCLE (right).

in μ_i itself rather than the error structure. This would be consistent with the extended Freundlich curve being an empirical approximation that may not perfectly capture the sorption mechanism across the full covariate range (Sibbesen, 1981). With $n = 23$ this cannot be formally tested, but it is a limitation worth acknowledging.

(f) Comparison of Small-Scale Component

Table 15: Comparison of small-scale parameter estimates across methods.

Quantity	Q2 Optimal EE	Q3 MLE	Q4 MCLE
θ	0.5422	0.5598	0.5752
σ^2	0.6573	0.6250	0.6437

Table 16: All estimates of σ^2 across methods.

Estimator	$\hat{\sigma}^2$
Q1 GLS (marginal variance)	1.0149
Q2 moment-based	0.6573
Q3 MLE	0.6250
Q4 MCLE	0.6437

The Q4 estimates of θ and σ^2 are close to Q3, as expected given that the two likelihoods differ by exactly one term. The small monotone progression in $\hat{\theta}$ ($0.542 \to 0.560 \to 0.575$ across Q2, Q3, Q4) is consistent with successive reduction of first-stage approximation error: Q2 uses residuals from misspecified large-scale parameters, Q3 jointly estimates all parameters from the full conditional likelihood, and Q4 does the same from the pairwise conditional likelihood. The σ^2 estimates follow the same hierarchy noted in Q2(d) and Q3(f): Q1 GLS is substantially larger (1.015) because it targets the marginal variance, while Q2, Q3, and Q4 all target the innovation variance directly and are in close agreement (0.657, 0.625, 0.644).

Q5

A two-stage estimation strategy would consist of estimating α, β, γ from the nonlinear constant variance model of Q1, and then using the residuals from that fit to estimate θ and σ^2 via the optimal estimating equation of Q2. Based on what you have learned in completing this exam, what would you conclude about the possibility of using such a two-stage procedure to fit the overall extended Freundlich model to data from this problem?

Stage 1 is misspecified in a way that contaminates Stage 2

The first-stage GLS fit assumes $w_i \overset{iid}{\sim} N(0, \sigma^2)$. When the true errors follow an AR(1) process with $\theta \approx 0.56$, this misspecification has two compounding consequences.

First, the GLS estimating equations use weight matrix $W = I$ in place of the efficient $\Sigma^{-1}(\theta)$, producing a consistent but inefficient estimator of (α, β, γ) in large samples. The Q1 covariance matrix, which assumes $W = I$, is therefore incorrect even as a large-sample approximation, and the resulting standard errors are unreliable for inference regardless of what happens in Stage 2.

Second, and more critically, the Q1 raw residuals $\hat{e}_i = y_i - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$ are noisy surrogates for the true w_i, subject to systematic finite-sample distortion. The difference

$$\hat{e}_i - w_i = \mu_i(\alpha, \beta, \gamma) - \mu_i(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$$

is a smooth nonlinear function of the Stage 1 parameter estimation error. Because μ_i varies smoothly with the ordered covariate, runs of positive or negative estimation error across adjacent x_i values inflate or deflate the lag-1 autocorrelation of $\{\hat{e}_i\}$ relative to that of $\{w_i\}$. With $n = 23$ this contamination is not asymptotically negligible.

The numerical results confirm this distortion

The naive estimating equation applied to Q1 residuals returned $\tilde{\theta}_{\text{naive}} = 1.005$, outside the stationary parameter space $|\theta| < 1$. This is a symptom of the trend contamination described above, not a sampling anomaly: the systematic residual pattern visible in the Q1 panel of Figure 3 — positive residuals at low and high covariate values, a slight dip in the mid-range — is exactly the signature of a misspecified error structure, and the naive estimating function has no mechanism to separate genuine autocorrelation from it.

The optimal estimating equation fares better ($\tilde{\theta}_{\text{opt}} = 0.542$) because multiplying by W_{i-1} down-weights large residuals and focuses on local lag-1 relationships, making it more robust to the smooth global trend. Nevertheless $\tilde{\theta}_{\text{opt}} = 0.542$ remains below the Q3 MLE of 0.560: the smooth trend in $\hat{e}_i - w_i$ reduces the apparent lag-1 dependence, attenuating the autocorrelation estimate toward zero relative to its true value.

Efficiency loss relative to the joint MLE

Table 17: Small-scale estimates and mean confidence band half-widths across all methods.

Method	$\hat{\theta}$	$\hat{\sigma}^2$	Mean Confidence Band Half Width
Q1 GLS	—	1.015	0.623
Q2 Optimal EE	0.542	0.657	—
Q3 MLE	0.56	0.625	0.967
Q4 MCLE	0.575	0.644	1.028

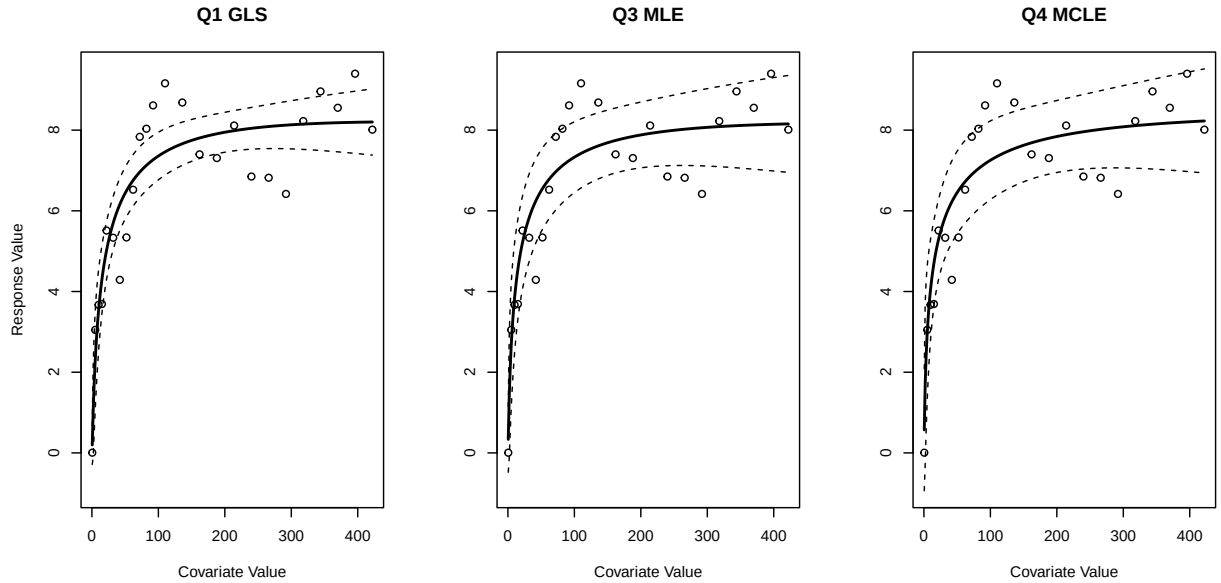

Figure 5: Fitted extended Freundlich curves with approximate 95% pointwise confidence bands: Q1 GLS (left), Q3 MLE (center), Q4 MCLE/Godambe (right).

As established in Q2(d), Q1 GLS targets the marginal variance $\sigma^2/(1-\theta^2)$ while all remaining methods target the innovation variance σ^2 directly.

Beyond the finite-sample distortion already described, efficiency is lost relative to the joint MLE even asymptotically. The Q3 MLE shifts $\hat{\alpha}$ from 0.506 to 0.739 and $\hat{\beta}$ from 1.222 to 1.028 — non-trivial movements that reflect the reweighting of observations once the AR(1) structure is correctly incorporated. This loss is a meaningful finite-sample concern: because the likelihood couples the regression and AR(1) parameters, the two-stage estimator with a fixed $W = I$ weight matrix is inefficient. In large samples, a *feasible* GLS procedure that first estimates θ consistently and then uses $\hat{\Sigma}^{-1}(\hat{\theta})$ as the weight matrix can recover the asymptotic efficiency of the joint MLE under standard regularity conditions—the two-stage procedure as described here does not do this, because Stage 1 fixes $W = I$ rather than iterating. The efficiency loss is therefore a consequence of the specific implementation, not an inherent property of decoupled estimation.

Figure 5 and Table 17 make the practical consequence visible across all methods simultaneously. The Q1 confidence band is the narrowest of the three, but this narrowness is illusory — it reflects the iid assumption forcing underestimation of uncertainty rather than genuine precision. The Q3 and Q4 bands are wider throughout, and correctly so: they acknowledge the curvature of the five-parameter likelihood surface and the correlation structure of the errors jointly. The mean band half-widths in Table 17 (0.623 for Q1, 0.967 for Q3, 1.028 for Q4) quantify this directly. The Q4 band is the widest because the Godambe sandwich covariance, even approximately, propagates more uncertainty than the inverse Hessian alone. Taken together, Figure 5 and Table 17 provide the clearest single summary of why the two-stage procedure understates inferential uncertainty: its narrow band is not an achievement but a consequence of ignoring what Q3 and Q4 explicitly model.

The Q4 composite likelihood provides corroborating evidence

The Q4 MCLE is close to Q3 ($\hat{\theta} = 0.575$, $\hat{\sigma}^2 = 0.644$), confirming that adjacent pairs carry most of the information about θ and that Q3 and Q4 are in close numerical agreement throughout. But constructing even approximately valid interval estimates for Q4 required invoking martingale difference sequence theory outside the course syllabus, and the Godambe-based intervals carry a formal caveat at $n = 23$. This is

itself informative: the inferential difficulty that Q4 makes explicit is the same difficulty that the two-stage procedure sidesteps by ignoring the correlation structure entirely.

Conclusion

The two-stage procedure fails for compounding reasons: a misspecified Stage 1 weight matrix, systematically distorted residuals passed to Stage 2, and an efficiency loss that persists even asymptotically because the likelihood couples the regression and AR(1) parameters. The numerical evidence is consistent: $\tilde{\theta}_{\text{naive}} > 1$ signals trend contamination, $\tilde{\theta}_{\text{opt}}$ is attenuated toward zero, and the Q1 confidence band is artificially narrow. Figure 4 and Table 18 bring the full picture into focus.

Among the estimators considered, the joint MLE of Q3 is the most defensible. For formal inference at this sample size, a parametric bootstrap calibrated to the Q3 model would be the natural next step.